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In [3]: from scipy.integrate import odeint
import numpy as np
import matplotlib.pyplot as plt

def lorentz(variables, t, p, s, b):
    x, y, z = variables
    dxdt = s * (y - x)
    dydt = x * (p - z) - y
    dzdt = x * y - b * z
    return [dxdt, dydt, dzdt]

t = np.linspace(0, 30, 10000)

init1 = [0.0, 1.0, 1.0]
init2 = [0.01, 1.0, 1.0]
init3 = [0.02, 1.0, 1.0]

sol1 = odeint(lorentz, init1, t, args=(70, 10, 8/3))
sol2 = odeint(lorentz, init2, t, args=(70, 10, 8/3))
sol3 = odeint(lorentz, init3, t, args=(70, 10, 8/3))

x1 = sol1[:, 0]
x2 = sol2[:, 0]
x3 = sol3[:, 0]

plt.figure(figsize=(15, 6))
plt.plot(t, x1, label='init1: [0.0, 1.0, 1.0]', color='red')
plt.plot(t, x2, label='init2: [0.01, 1.0, 1.0]', color='blue')
plt.plot(t, x3, label='init3: [0.02, 1.0, 1.0]', color='green')
plt.title('x(t) for Varying Initial Conditions - 09:59 PM +0530, Tuesday, August 05')
plt.xlabel('t')
plt.ylabel('x')
plt.legend()
plt.grid(True)
plt.show()
```



